

Task 01

1. Your company has the network 192.168.50.0/24 and needs 6 subnets.
How would you design it?

Network: 192.168.50.0/24

Subnet mask: 255.255.255.0

Total IPs: 256

Usable hosts: 254

Need at least 6 subnets

$2^3 = 8 \geq 6 \rightarrow$ So we borrow 3 bits

New subnet mask: /24 + 3 = /27

Host bits = 32 - 27 = 5

Total IPs = $2^5 = 32$

Usable hosts = 32 - 2 = **30 hosts per subnet**

Subnet	Network Address	First Host	Last Host	Broadcast Address
1	192.168.50.0/27	192.168.50.1	192.168.50.30	192.168.50.31
2	192.168.50.32/27	192.168.50.33	192.168.50.62	192.168.50.63
3	192.168.50.64/27	192.168.50.65	192.168.50.94	192.168.50.95
4	192.168.50.96/27	192.168.50.97	192.168.50.126	192.168.50.127
5	192.168.50.128/27	192.168.50.129	192.168.50.158	192.168.50.159
6	192.168.50.160/27	192.168.50.161	192.168.50.190	192.168.50.191

2. You are given a subnet 172.20.0.0/20. How many usable hosts can each subnet provide?

CIDR: /20

Subnet mask: 255.255.240.0

Host bits = 32 - 20 = 12

Total IPs: $2^{12} = 4096$

Usable Hosts: 4096 - 2 = **4094 usable hosts**

Subnet 172.20.0.0/20 provides 4094 usable hosts